

# **AI-GENERATED MODEL PAPER**

Subject: Database Management Systems

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*Total Marks: 100*

# Examination Paper

## Question Q1 (20 marks)

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- a.i) Explain the process of logical database design and its importance in the overall database design process. (4)
- a.ii) Describe the different types of integrity constraints (domain, referential, key, other) and provide an example for each type. (4)
- a.iii) Explain how to convert a complex ER model to the relational model and provide a step-by-step guide on this process. (5)
- b.) Given an ER diagram with entities 'Employee', 'Department', and 'Project', calculate the primary keys for each entity and explain how they relate to the overall database schema. (Calculate type) (7)

## Question Q2 (20 marks)

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- a)) Draw an entity-relationship diagram for the following scenario: A library has several sections such as fiction, children and technology. Each section has a name and a number(unique) and its headed by a head librarian. Each book belong to a section and has a title, authors, ISBN, year and a publisher. A book may have several copies. Each copy is identified by an access number. For each copy borrowed, current borrower and due date should be tracked. Members have a membership number, an address and a phone number. Members can borrow 5 books and could put hold request on a book Librarian has a name, id number, phone and an address. (2)
- b)) List the four normal forms of relational databases: First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF) and Boyce-Codd Normal Form (BCNF). Explain each in detail. (2)
- c)) A relation R has the following set of functional dependencies over R:  $\{A \rightarrow B, A \rightarrow C, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$ . Compute the keys for relation R. Explain your reasoning. (2)
- d)) Explain the concept of superkey and key in the context of relational databases. Provide examples to illustrate each. (2)
- e)) A relation R has the following set of functional dependencies over R:  $\{A \rightarrow B, A \rightarrow C, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$ . Determine if the relation is in Boyce-Codd Normal Form (BCNF). Explain your reasoning. (5)

f)) Explain the concept of transitive dependency and its implications on normalization. Provide examples to illustrate each. (2)

g).i.) A relation R has the following set of functional dependencies over R:  $\{A \rightarrow B, A \rightarrow C, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$ . Normalize the relation to its highest normal form. Explain your reasoning. (2)

g).ii.) Consider a relation R with the following set of functional dependencies over R:  $\{A \rightarrow B, A \rightarrow C, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$ . Determine if the relation is in Boyce-Codd Normal Form (BCNF). Explain your reasoning. (3)

### Question Q3 (22 marks)

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Part a: General type) Explain the concept of EER model and its significance in database design. (2)

Part b: General type) Describe the difference between ternary and aggregation relationships in EER diagrams. (2)

Part c: Define type) Define specialization and generalization concepts in the context of EER model. (2)

Part d: Define type) Explain overlapping constraint and its application in EER diagrams. (3)

Part e.i: General type) Draw an ER diagram for the given scenario: A Library is organized into several sections such as fiction, children and technology. Each section has a name and a number(unique) and its headed by a head librarian. (2)

Part e.ii: General type) Explain the concept of fan trap in EER diagrams and provide an example. (3)

Part e.iii: General type) Explain the concept of chasm trap in EER diagrams and provide an example. (2)

Part e.iv: General type) Describe the process of logical database design and its importance in database management systems. (3)

Part e.v: General type) Explain the concept of relational model and its advantages over other data models. (3)

### Question Q4 (20 marks)

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a) What is the primary key for the Students relation in the given ER model? (2)

- b) Explain the concept of referential integrity constraint and its application in the given ER model. (3)
- c.i) Draw an EER diagram for the Students relation, including the primary key and foreign keys. (4)
- c.ii) Explain the concept of aggregation in ER modeling and provide an example from the given scenario. (2)
- d.i) Identify the entities that participate in a fan trap relationship in the given ER model. (2)
- d.ii) Design an EER diagram for the given scenario, including all relationships and constraints. (7)

### Question Q5 (18 marks)

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- a) Draw an ER diagram for a university database that stores information about courses, lecturers, and their qualifications. The database should include the following entities: Lecturer, Qualifications, Course, and Teach. Ensure that the ER diagram accurately represents the relationships between these entities. (18)
- b) Explain the concept of normalization in database design and provide an example of a normalized database schema for the university database. Be sure to include the benefits of normalization. (0)